

Programme	BSc Statistics				
Course Code	STA2MN106 (P)				
Course Title	Correlation and regression				
Type of Course	Minor				
Semester	II				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Understanding of fundamental statistical concepts, familiarity with common data formats and basic data processing.				
Course Summary	Delve into advanced statistical techniques like skewness, kurtosis, multiple correlation, multiple regression, and R programming, equipping students to apply statistical analysis practically in real-world scenarios.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Grasp skewness, kurtosis, and their measures.	U	C	Instructor-created exams / Quiz
CO2	Define correlation and distinguish between positive, negative and zero correlation and critically evaluate ethical implications of statistical methods aligning with human values.	U	C	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Understand partial and multiple correlation and analyze data to help entrepreneurial decisions using critical thinking skills.	U	F	Seminar Presentation / Group Tutorial/ Instructor-created exams Work
CO4	Describe and apply multiple linear regression to model relationship with more than one predictor variable.	U	C	Instructor-created exams / Home Assignments/ Instructor-created exams
CO5	Implement multiple regression techniques effectively.	U	C	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Introduce and apply R programming for	Ap	P	Viva Voce/

	statistical analysis.			Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hours (45 +30)	Marks 70
I		Skewness and Kurtosis	9	15
	1	Skewness	2	
	2	Kurtosis	2	
	3	Pearson's measure of skewness	3	
	4	Percentile measure of Kurtosis	2	
		Unit 1:3.13 Ref[2] Unit 2: 3.14 Ref[2] Unit 3: 3.13 Ref[2] Unit 4: 3.14 Ref[2]		
II		Partial and multiple correlation	14	20
	5	Partial correlation	2	
	6	Partial correlation coefficient	2	
	7	The Significance of a Partial Correlation Coefficient	2	
	8	Multiple correlation	2	
	9	Coefficient of Multiple Correlation	2	
	10	Advantages of Multiple Correlation Analysis	2	
	11	Limitations of Multiple Correlation Analysis	2	

	Unit 5: 9.2 Ref[1] Unit 6: 9.2 Ref[1] Unit 7: 9.6 Ref[1] Unit 8: 9.7Ref[1] Unit 9: 9.7 Ref[1] Unit 10: 9.8 Ref[1] Unit 11: 9.9 Ref[1]			
III	Multiple regression		12	20
	12	Multiple regression	3	
	13	Multiple Regression and Correlation Analysis	3	
	14	Assumptions of Linear Multiple Regression Analysis	3	
	15	Coefficient of Multiple Determination	3	
	Unit 12: 9.9 Ref[1] Unit 13: 9.9 Ref[1] Unit 14: 9.10 Ref[1] Unit 15: 9.7 Ref[1]			
IV	Introduction to R programming		10	15
	16	Installing R	1	
	17	Objects in R	1	
	18	Using functions in R	1	
	19	Importing data	1	
	20	Exporting data	1	
	21	Simple base R plots	2	
	22	Multiple graphs	3	

	Unit 16: 1.1 Ref[3] Unit 17: 2.2 Ref[3] Unit 18:2.3 Ref[3] Unit 19: 3.3 Ref[3] Unit 20: 3.6 Ref[3] Unit 21: 4.2 Ref[3] Unit 22: 4.4 Ref[3]		
V	PRACTIUM	30	
	Do practice problems using R software from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.		
	1 Correlation in R		
	2 Customising plots		
	3 Simple base r plots		
	4 R packages		
	5 Installing R studio		
	6 Projects in R studio		
	7 Backing up projects		
	8 File names		
	Sections from References: Unit 1: 6.2 Ref[3] Unit 2: 4.3 Ref[3] Unit 3: 4.2 Ref[3] Unit 4: 1.5Ref[3] Unit 5: 1.2 Ref[3]		

Unit 6: 1.6 Ref[3]		
Unit 7: 1.12 Ref[3]		
Unit 8: 1.9 Ref[3]		
Books and References: 1. S.P Gupta (2021), Statistical Methods 46 th Edition 2. Gupta, S.C. and Kapoor, V.K. (1997) Fundamentals of Mathematical Statistics. Sultan Chand and Sons, New Delhi 3. Douglas, Alex, Deon Roos, Francesca Mancini, Ana Couto, and David Lusseau. (2020), <i>An Introduction to R</i>. https://intro2r.com/index.html.		

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	3	-	-	-	3	1	2	-	2	2	-
CO 2	1	-	2	-	-	3	2	1	-	2	2	3
CO 3	-	-	2	-	-	2	2	2	-	1	3	-
CO 4	-	-	-	-	-	-	1	1	-	2	-	-
CO 5	-	3	-	2	2	-	3	2	2	2	-	-
CO 6	-	-	-	-	-	2	2	1	-	2	-	-

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial /

	High
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Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4	✓	✓		✓
CO 5		✓		✓
CO 6	✓			

Programme	BSc Statistics
Course Code	STA3MN206 (P)